Applications of Pythagoras' theorem

Core Concepts: Applications of Pythagoras' theorem

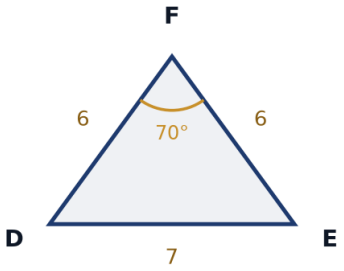
1 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 97°

b. 107°

c. 89°

d. 142°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 16

c. 14

d. 19

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 142°

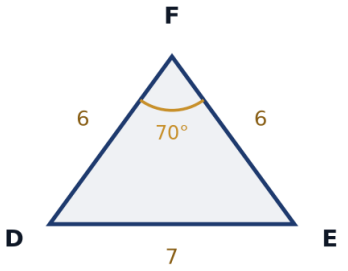
5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm

c. 28 cm

d. 63 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 14

b. 21

c. 19

d. 24

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 83°

b. 93°

c. 75°

d. 142°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

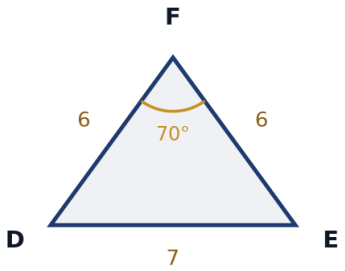
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 19

b. 26

c. 24

d. 29



Exam-Style Problem Solving & Modeling

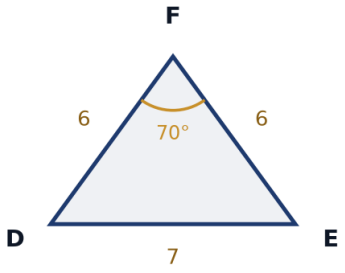
10 In $\triangle ABC$ studying applications of pythagoras' theorem, $m\angle A = 38^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 76°

b. 86°

c. 68°

d. 142°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29

d. 34

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